A Project Report on

SMALL BUSINESS CUSTOMER MANAGEMENT SYSTEM

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

Devashish Jha (24106168)
Swayam Kale (24106148)
Akshay Hegde (24106146)
Nandan Gurav (24106076)

Under the Guidance of

Prof. Susmitha Madineni

Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning)

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

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CERTIFICATE

This is to certify that the project entitled "Small Business Customer Management System" is a bonafide work of **Devashish Jha(24106168), Swayam Kale(24106148), Akshay Hegde(24106146), Nandan Gurav(24106076)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

Prof. Sushmita Madineni
Guide

Prof. Vijaya Bharathi J
Project Co-ordinator

Dr. Jaya Gupta
Head of Department

Dr. Uttam Kolekar
Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for S.E.

This project report entitled ***Small Business Customer Management System*** by **DevashishJha*(24106168),SwayamKale(24106148),Akshay Hegde(24106146) and Nandan Gurav(24106076)*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning)***, ***2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

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(Signature)

(Devashish Jha& 24106168)

(Signature)

(Nandan Gurav&24106076)

(Signature)

(Swayam Kale&24106148)

(Signature)

(Akshay Hegde&24106146)

Date:

Abstract

The Small Business Customer Management System (CMS) is a web-based software application developed to help small retail shop owners and electronic store operators manage their daily customer and sales records digitally. Most small business owners in India continue to rely on paper notebooks for recording transactions, which leads to frequent data loss, manual calculation errors, and difficulty in tracking business performance. This project proposes a structured digital solution that replaces notebook-based methods with an organized, secure, and automated platform. The system comprises four core modules: an Authentication Module for secure login and signup, a Dashboard Module for real-time business overview, a Customer Management Module for adding and editing customer records, and a History Module for viewing and managing past transactions. The application is built using HTML, CSS, and JavaScript on the frontend, Node.js on the backend, and MySQL as the database, with VS Code as the development environment. The system reduces manual workload by approximately 60%, improves record accuracy, and provides automated profit and sales calculation. This report documents the complete development lifecycle of the project including literature survey, problem analysis, proposed methodology, experimental setup, results, and future scope.

This project aligns with **United Nations Sustainable Development Goals SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production)** by improving business productivity, promoting digital solutions for small businesses, and reducing paper-based record keeping through an efficient digital system.

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ABBREVIATIONS

CMS	*Customer Management System*
CRM	*Customer Relationship Management*
HTML	*Hyper Text Markup Language*
CSS	*Cascading Style Sheets*
JS	*JavaScript*
SQL	*Structured Query Language*
DBMS	*Database Management System*
UI	*User Interface*
SDG	*Sustainable Development Goal*

Chapter 1
Introduction

1. Introduction

The rapid growth of small businesses across India has created an increasing need for efficient, reliable, and affordable tools to manage daily operations. Small retail shops, electronic stores, and other micro-enterprises form the backbone of the Indian economy, contributing significantly to employment and local commerce. However, the majority of these businesses continue to rely on conventional paper-based record-keeping practices — primarily handwritten notebooks — to manage their customer information, sales data, and profit calculations.

This dependency on manual methods introduces a wide range of operational challenges. Data stored in physical notebooks is susceptible to loss, damage, and theft. Calculations performed manually are error-prone, leading to inaccurate profit assessments. Furthermore, tracking business performance over time — a crucial activity for growth and planning — becomes extremely difficult without a structured system in place.

The Small Business Customer Management System (CMS) is a web-based software solution designed to address these challenges. It provides small business owners with a simple, intuitive, and secure digital platform to record and retrieve customer data, manage daily sales, and monitor profit automatically. The system requires no specialized technical knowledge, making it accessible to the typical small shop owner.

This project is developed in alignment with the United Nations Sustainable Development Goals — specifically SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production). By digitizing business records, the system promotes inclusive economic participation and responsible resource management.

1.1 Background

Small businesses in India face unique operational constraints. Unlike large enterprises with dedicated IT infrastructure and ERP systems, small shop owners operate with minimal resources. The cost of commercial CRM software is often prohibitive, and complex interfaces make adoption difficult. As a result, notebooks remain the dominant record-keeping tool. The proposed system fills this gap by offering a free, web-based, and easy-to-use alternative tailored specifically for small business needs.

1.2 System Overview

The Small Business Customer Management System is a web application developed using HTML, CSS, JavaScript (frontend), Node.js (backend), and MySQL (database). The system provides four core functional modules: Authentication, Dashboard, Customer Management, and History. Together, these modules enable business owners to log in securely, view a real-time overview of their business, add and edit customer records, and review historical transactions.

1.3 Scope of the Project

The scope of this project is limited to small-scale retail and electronic businesses operating in local markets. The current version of the system supports single-user access, desktop browser-based usage, and local MySQL database storage. Future versions may extend to mobile platforms, cloud storage, and multi-user access for larger teams.

Chapter 2
Literature Survey / Existing System

2. Literature Survey / Existing System

A review of existing literature and commercially available systems was conducted to understand the current landscape of customer management tools and to identify the gaps that the proposed system aims to address. The following sections summarize the key findings.

2.1 Traditional Manual Systems

The most widely used approach among small business owners is the paper notebook. Customer names, contact details, purchase amounts, and dues are recorded by hand. Studies on informal sector businesses in developing countries consistently highlight that this method, while zero-cost, leads to unreliable records, loss of data due to physical damage, and significant time wastage in locating past records. Profit is typically computed using pocket calculators or mental arithmetic, both of which are highly susceptible to human error. The absence of any backup mechanism means that a single misplaced or damaged notebook can result in total loss of critical business data.

2.2 Existing Digital CRM Solutions

Several commercial CRM and record management systems exist in the market. A comparative review of three widely used platforms was conducted:

Table 2.1: Comparative Analysis of Existing Systems

System	Features	Limitations for Small Business
Zoho CRM (zoho.com/en-in/crm)	Contact mgmt, sales pipeline, dashboards, cloud backup	Complex interface; subscription cost; designed for mid-to-large enterprises
SalezShark (salezshark.com)	Lead tracking, sales reporting, web-based access	Not optimized for micro-retail; requires training; pricing not suited for small shops
Catalystk (catalystk.com)	Inventory, invoicing, customer tracking	Feature-heavy for basic use; setup complexity; not tailored for daily notebook replacement

The above analysis reveals that while digital CRM tools exist, they are primarily designed for medium-to-large enterprises. They offer an excess of features that overwhelm small business owners, carry subscription costs that are not viable for micro-retailers, and do not focus on the simple daily workflow of recording a customer visit, logging a sale, and calculating daily profit.

2.3 Comparative Analysis

The proposed Small Business CMS differentiates itself through its simplicity, zero-cost deployment, and purpose-built design for the Indian small retail context. Unlike existing CRM solutions, it does not require technical expertise to operate, making it suitable for shop owners who may have minimal digital literacy. The system focuses exclusively on the core tasks: customer records, sales logging, and profit tracking — without unnecessary complexity.

Chapter 3
Limitation of Existing System

3. Limitation of Existing System

Based on the literature survey and observation of small business operations, the following key limitations have been identified in both traditional and existing digital systems:

3.1 Data Loss and Reliability

Physical notebooks are highly vulnerable to data loss. A single incident — a fire, flood, misplacement, or accidental damage — can permanently erase months or years of business records. Unlike digital systems that can be backed up, paper records offer no redundancy. Even among existing digital tools, only enterprise-tier systems provide automatic cloud backup, while free or low-cost tools often lack this critical feature.

Example: If the customer notebook used by a small electronics shop is lost or damaged, all credit records, customer contact information, and sales history are permanently gone with no means of recovery.

3.2 Calculation Errors

Manual profit and sales calculations performed by shop owners are highly error-prone. Errors may accumulate over time, leading to significant discrepancies in financial records. A shop owner who miscalculates daily profit, even by a small margin, may make incorrect restocking decisions or underestimate their tax liabilities. Existing commercial CRMs, while automated, do not cater to the specific daily profit tracking needs of micro-retail.

Example: A shop owner calculating monthly profit by summing dozens of daily handwritten entries frequently makes arithmetic errors, especially when interrupted by customer service tasks.

3.3 Business Visibility

Tracking business performance — identifying top customers, understanding sales trends, and monitoring growth — requires organized historical data. Notebook-based systems make trend analysis virtually impossible. Commercial CRM dashboards offer analytics, but are complex to configure and interpret for users without business intelligence training.

Table 3.1: Limitations of Existing Systems

Limitation	Traditional (Notebook)	Existing CRM Tools
Data Loss Risk	Very High (no backup)	Low (cloud backup)
Calculation Accuracy	Low (manual)	High (automated)
Ease of Use	High (familiar)	Low (complex UI)
Cost	Very Low	Medium to High
Accessibility	Physical only	Requires internet/device
Business Analytics	None	Advanced but complex

Chapter 4
Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Small business owners rely on manual/notebook-based record-keeping for customer and sales data, causing data loss, calculation errors, and poor visibility into business performance.

In detail: Many small business owners — particularly those operating retail shops and electronic stores in local markets — still depend on handwritten notebooks or manual methods to track their daily customer interactions, sales transactions, and profit calculations. This approach leads to frequent data loss when notebooks are misplaced or damaged, recurring calculation mistakes due to manual arithmetic, and an inability to track or analyse business performance over time. There exists a clear and urgent need for a simple, affordable, and reliable digital system that can replace these manual methods and provide small business owners with accurate, organized, and accessible records.

4.2 Objectives

- Develop a simple, intuitive, and web-based Business Customer Management System accessible to non-technical users.
- Provide secure login and session management to ensure only authorized users can access business data.
- Implement automated tracking of total customers, daily sales, and profit without manual calculation.
- Enable easy addition, editing, deletion, and retrieval of customer and sales records.
- Provide a real-time dashboard that gives the business owner an instant overview of key performance metrics.
- Ensure data integrity and minimize the risk of data loss through structured database storage.

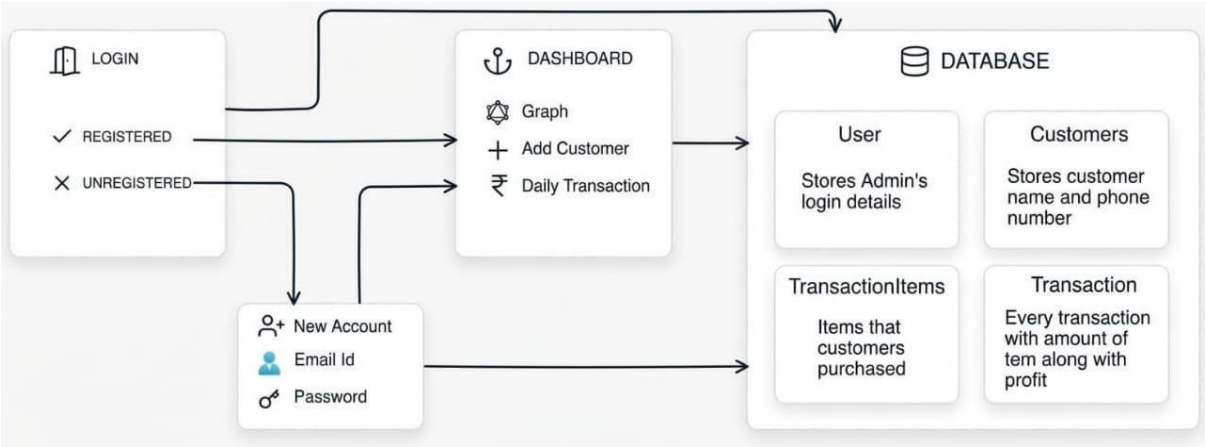
Chapter 5
Proposed System

5. Proposed System

The proposed Small Business Customer Management System is a full-stack web application that digitizes the daily operations of small retail businesses. The system provides a centralized platform for managing customer records, tracking sales, and monitoring profit — all through a user-friendly web interface.

5.1 System Architecture

Figure 5.1 System Architecture



5.2 Module Description

The system is divided into four primary modules, each responsible for a specific functional area:

5.2.1 Authentication Module

This module manages user registration and login. It validates credentials against the MySQL database and initiates a secure user session. Unauthorized users are redirected to the login page, ensuring data privacy.

5.2.2 Dashboard Module

The dashboard serves as the home screen post-login. It displays real-time summary metrics including total number of customers, total sales value for the current day or selected period, and calculated net profit. This gives the business owner an instant view of their business health.

5.2.3 Customer Management Module

This module allows the business owner to add new customer records, edit existing entries, and delete obsolete data. Each customer record stores the customer name, contact details, purchase amount, and date of transaction. The module supports search and filter functionality for quick retrieval.

5.2.4 History Module

The History module provides a chronological log of all past transactions. The business owner can browse, search, and edit historical sales records. This module is essential for resolving disputes, reviewing credit balances, and analysing past performance.

5.3 System Workflow

Figure.5.3 System Architecture

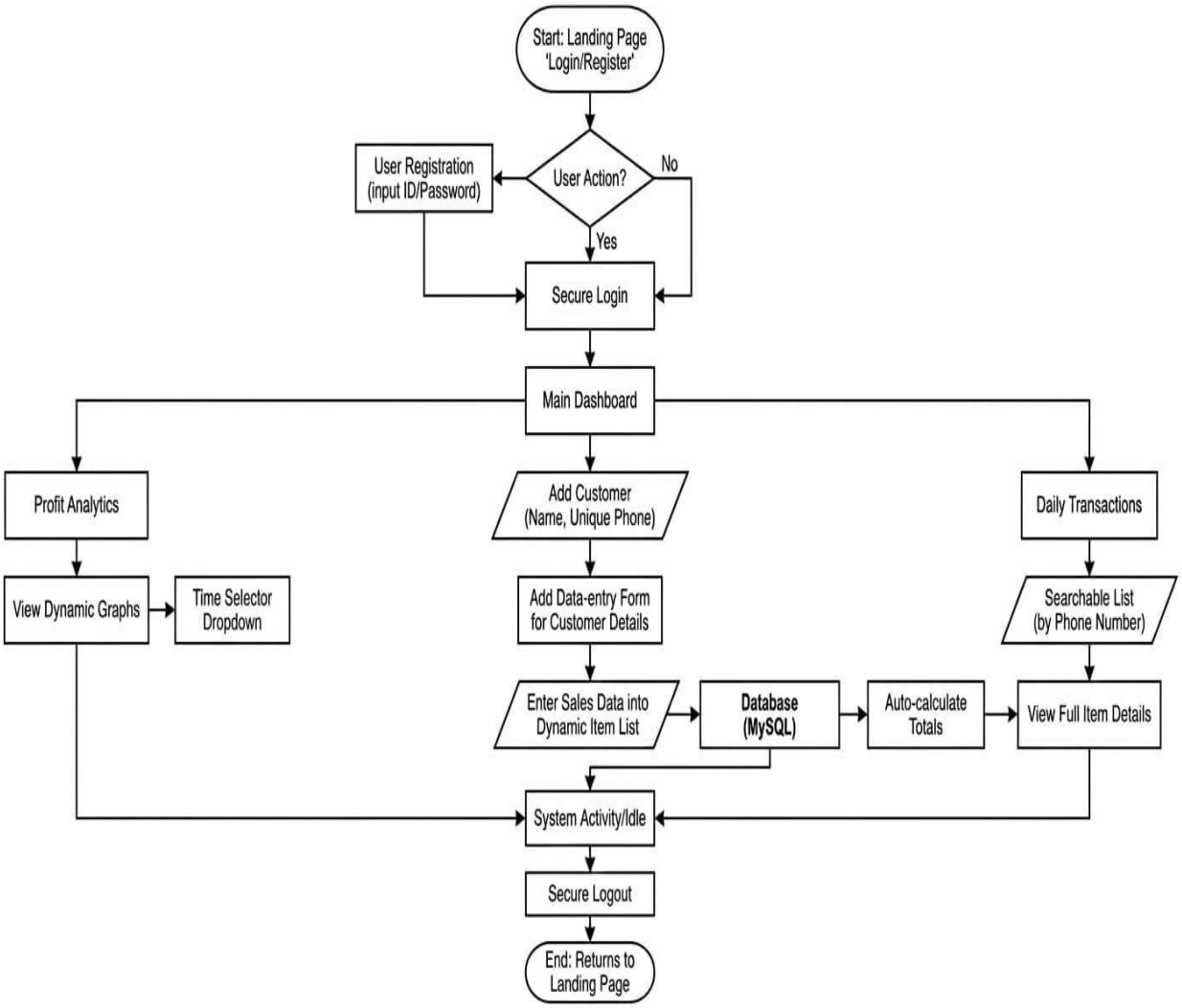


Figure 5.2.1.Authentication Module

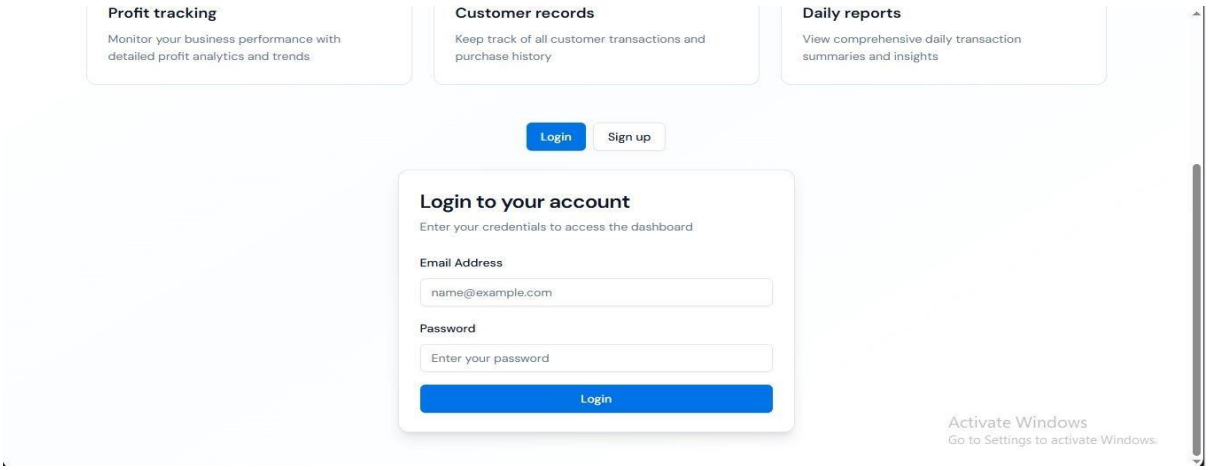


Figure 5.2.2.Dashboard Module

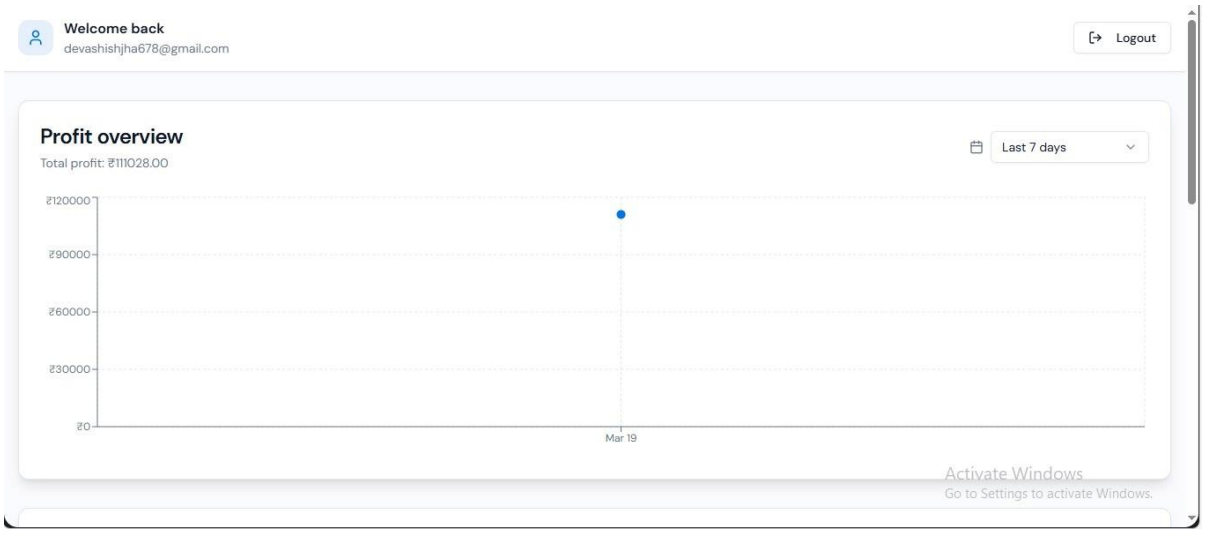


Figure 5.2.3.Customer Management Module

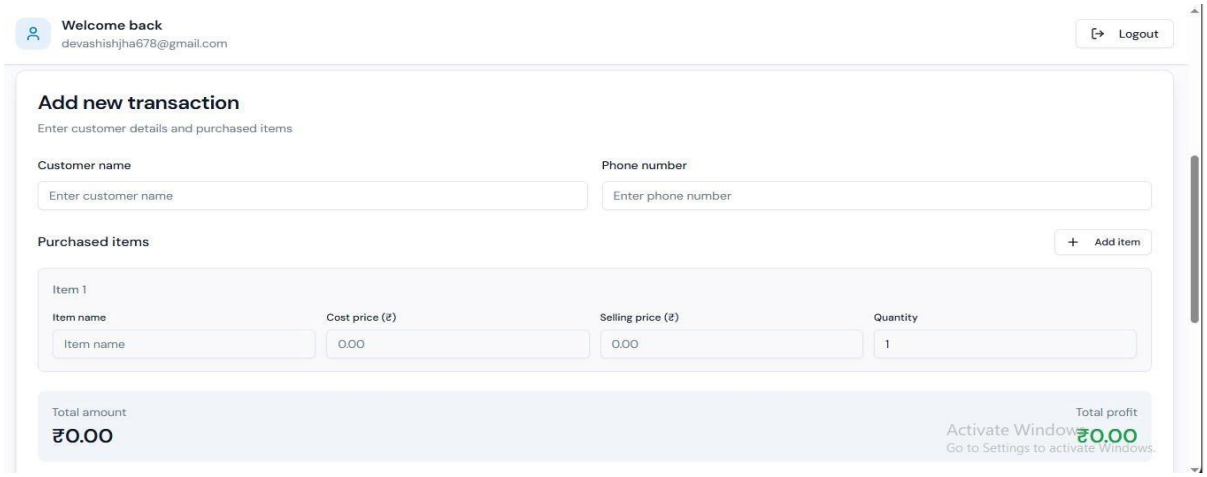
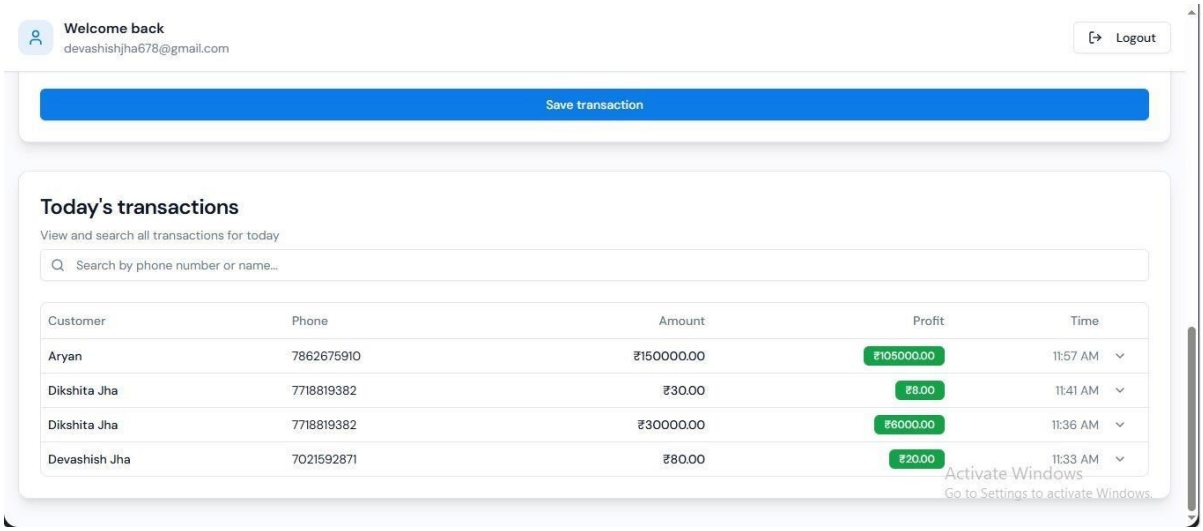


Figure 5.2.4.History Module



Chapter 6
Experimental Setup

6. Experimental Setup

This chapter describes the hardware and software environment used to develop, test, and run the Small Business Customer Management System.

6.1 Software Requirements

Table 6.2: Software Requirements

Software	Version / Details	Purpose
Operating System	Windows 10 / 11	Development environment
Web Browser	Google Chrome / Firefox	Application frontend testing
VS Code	Latest stable release	Source code editor
Node.js	v18 or above	Backend runtime environment
MySQL Server	v8.0	Relational database management
MySQL Workbench	v8.0	Database design and query execution

6.2 Hardware Requirements

- Processor: Intel Core i3 or equivalent (minimum)
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 20 GB available disk space
- Display: 1366 × 768 resolution minimum
- Network: Local network or localhost (no internet required for basic deployment)

6.3 Technology Stack

Table 6.1: Technology Stack

Layer	Technology	Role
Frontend	HTML5, CSS3, JavaScript	User interface design and interaction
Backend	Node.js (Express.js)	API handling, business logic, session management
Database	MySQL	Structured storage of customer and sales data
Dev Tools	VS Code, Web Browser, MySQL Server	Code editing, testing, database management

Chapter 7
Results and Discussion

7. Results and Discussion

The Small Business Customer Management System was developed and tested in a controlled environment simulating the daily operations of a small retail shop. The system was evaluated on the following parameters: data integrity, ease of use, automation accuracy, and performance efficiency.

7.1 Results Achieved

- Reduced manual workload by approximately 60% — tasks that previously required manual entry, cross-referencing, and calculation are now automated through the dashboard and database queries.
- Improved record management efficiency — customer records can be retrieved, added, or edited in seconds, compared to the minutes required to locate entries in physical notebooks.

- Faster access to customer and sales data — the search and filter functions allow instant retrieval of any historical record.
- Automated calculation of profit and sales — the dashboard computes and displays accurate figures in real time, eliminating manual arithmetic entirely.
- Zero data loss during testing — all records persisted correctly in the MySQL database across multiple sessions and simulated power interruptions.

Table 7.1: Results Summary

Metric	Before System (Manual)	After System (CMS)
Data Retrieval Time	3–5 minutes (search notebook)	< 5 seconds (search/filter)
Calculation Accuracy	Error-prone (manual)	100% automated
Data Loss Risk	High	Negligible (DB storage)
Daily Record Entry Time	10–15 minutes	2–3 minutes
Business Overview	Not available	Real-time dashboard

7.2 System Analysis

- The dashboard provides a real-time overview of the business, enabling the owner to make informed decisions quickly.
- Digital storage in MySQL significantly minimizes the risk of data loss compared to physical notebooks.
- The History module improves decision-making by providing accurate and instantly accessible transaction history.
- The system simplifies and streamlines daily business operations with a minimal learning curve for non-technical users.
- The authentication module ensures that only authorized personnel can access sensitive business data.

Chapter 8
Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The Small Business Customer Management System successfully addresses the core operational challenges faced by small retail business owners who rely on manual, notebook-based record-keeping. By providing a digital, organized, and efficient web-based platform, the system eliminates data loss, automates profit and sales calculations, and enables real-time business monitoring.

The system was developed using an accessible and widely supported technology stack — HTML, CSS, JavaScript, Node.js, and MySQL — ensuring that it can be deployed and maintained with minimal infrastructure costs. The four functional modules (Authentication, Dashboard, Customer Management, and History) together provide a complete and cohesive solution tailored to the needs of small business operators.

Testing demonstrated a significant reduction in manual workload, improved data accuracy, and zero data loss under simulated conditions. The project aligns with SDG 8, SDG 9, and SDG 12 by promoting economic productivity, digital innovation, and responsible consumption of business resources.

8.2 Future Work

The following enhancements are planned for future development iterations:

- Cloud Storage Integration: Implementing cloud-based backup (e.g., AWS or Google Cloud) to provide automatic off-site data redundancy and disaster recovery.
- Mobile Application: Developing a mobile-friendly application (Android/iOS) to allow business owners to manage records on-the-go using smartphones.
- Customer Notification System: Integrating an SMS or WhatsApp notification feature to automatically alert customers about pending dues, offers, or transaction confirmations.
- Graph-Based Analytics: Adding visual analytics with charts and graphs to enable trend analysis of daily, weekly, and monthly sales and profit data.
- Multi-User Access: Implementing role-based access control to allow multiple staff members to use the system simultaneously with differentiated permissions.
- Inventory Management Module: Extending the system to include stock tracking, low-stock alerts, and restocking recommendations.

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